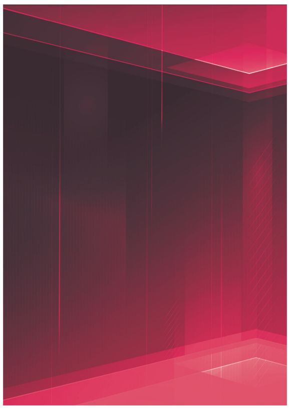

My testtt site

the greatest subtitle ever

tester

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2.1 Elastic Training Cluster

Elastic training has become a core technology in the deep learning field to address the computational demands of large-scale models and datasets. AI-Stack's "Elastic Training Cluster" feature is designed to help developers enhance the flexibility and efficiency of model training by distributing training workloads across multiple containers or nodes, effectively reducing training time.

Key Features and Characteristics

- Unified mounting environment

  Supports mounting users' Home directories, allowing containers to directly access development code, datasets, and scripts, improving workflow consistency.

- Support for mainstream training frameworks

  Supports distributed training frameworks such as Horovod and DeepSpeed, simplifying the configuration process for multi-machine collaborative training tasks.

- Dynamic resource scaling

  Container cluster resources can be elastically adjusted based on task requirements, supporting real-time scaling up or down of container quantities to improve resource utilization efficiency.

- User interface and monitoring

  Provides an intuitive interface to help users create, manage, and monitor training clusters, with built-in resource monitoring and performance analysis tools.

2.2 Creating Elastic Training Clusters

Before creating an elastic training container cluster, ensure the following information. If relevant operations need to be taken, please consult your administrator. For relevant configurations, please refer to the relevant sections in the Administrator Manual.

Before You Begin

1. Ensure that your project has access to available MLS Spec

- The specification must **not** include shared GPU mode
- Confirm the specification is available for your project

2. Ensure that the Public Image List provides suitable images

- Images compatible with DeepSpeed and Horovod frameworks are required
- If no suitable images are available, you can first use the container service to create a container, install the training framework yourself, and then create a custom image. Select the custom image with the training framework installed when creating the cluster.

> **Notice**
>
> 1. Containers created in the same batch must maintain consistent resource specifications. Different batches can be configured with different specifications, but it is recommended to maintain consistent specifications within the same cluster to ensure optimal performance and stability.
> 2. GPU resources are allocated in whole units (per GPU) for elastic training. Deploying training containers on nodes with shared GPU configurations is not supported.

2.2.1 Create Container Clusters

1. Access the Create Cluster Page

- Go to 【Container Management】 > 【Elastic Training Cluster】
- Click the `Create a Cluster` button

2. **Basic Settings**

- Select **Cluster Type** to create. Horovod and DeepSpeed frameworks are currently supported.
- Enter **Cluster Name** .
- Enter **Description** (Optional).
- Set **Schedule Time** ：
- **Start Time** : The default setting is Execute Immediately. You can also specify the date and time.
- **End Time** : The default setting is N/A. You can also specify the date and time or running hours.
- Click [Next] to configure Resources

3. **Resources**

- Set **Container Count** : The number must be greater than or equal to 2, which includes a launcher container and one or more worker containers.
- Launcher container: Responsible for unified configuration of container startup parameters, executing launch commands and managing communication flow
- Worker container: Handles the actual training tasks
- Select an **Image** with the training framework based on the cluster type.
- Configure the service **password** (e.g., SSH)
- **Enable GPU** (active by default)
- Select **Specification** based on your training needs. If you choose an Nvidia GPU, select CUDA Version

> **Note**
>
> After selecting a Specification, the Quota section at the top will display the quota usage of the selected specification. You can click to expand and view detailed GPU, vCPU, and RAM usage information, including current usage, quota limits, and remaining quota.

- Set Shared Memory (active by default, recommended to set a minimum capacity of 1 GB)
- Click [Next] to configure Storage

4. **Storage**

- Home storage:
- Configure mounting of **home storage** device, select the device to mount. If needed, you can enable [Default as Jupyter Workspace]
- If a new storage device needs to be added, click the [+] button on the right or [Create Storage] button

> **Home Storage**
>
> The platform mounts users' Home directories into the training environment. After mounting, files such as data generated during training, model cache, and personalized settings will be stored in this directory. When you need to find training-related data, you can search in the Home directory.
>
> > **Notice**
> >
> > a. Do not store large amounts of temporary files that don't have to be retained in the Home directory.
> >
> > b. Files under this directory will not be automatically deleted when a training cluster is removed.

- Internal storage:
- If needed, enable mounting of **internal storage** .
- Select the internal storage device to mount and configure the mounting path.

5. **Summary**

- Confirm all configurations. If you need to modify configurations, click the edit icon in the upper right corner to go back and edit.
- Click the [Submit] button

6. Eeturn to the 【Elastic Training Cluster】 page after submitting.

- When the status of the container cluster changes from `Creating` to `Created`, the elastic training cluster has been created successfully.

2.3 Elastic Training Cluster List

2.3.1 Cluster List

The 【Elastic Training Cluster】 page lists created container clusters, including the following information: - Cluster type

- Running status
- Launcher container count (created/total)
- Worker container count (created/total)
- Cluster start time

2.3.2 Cluster Details

Select the container cluster you want to view in the 【Elastic Training Cluster】 page to view the following details:

Details

View the following:

- Basic information of the container cluster
- Container count
- Selected image and MLS specification
- Storage device settings

Containers

View all containers and their running information within the cluster, including:

- Running status
- Start time
- GPU utilization
- GPU memory usage
- vCPU usage
- RAM usage

Monitoring

- View detailed monitoring data of a single container within the specified time:
- GPU utilization
- GPU memory usage
- vCPU usage
- RAM usage
- Network I/O
- Select the container you want to view from the drop-down menu. The specified time range (maximum query range is 30 days) can be viewed as well.

Services

- View all services provided by the cluster. The services and activation methods are configured by platform administrators using the MLS Templates (image). Supported services include:
- SSH
- Jupyter
- JupyterLab
- TensorBoard
- WebTerminal
- Code Server
- User-defined services

Logs

- View all log outputs generated during a single container's runtime
- Select the container you want to view from the drop-down menu

Events

- View key status changes or trigger events that occurred during a single container's runtime

- Select the container you want to view from the drop-down menu

2.3.3 Cluster Services

In the 【Elastic Training Cluster】 page, follow these steps to quickly access cluster services:

1. Select the cluster you want to view
2. Click the [Service] button to quickly access the service page for this container cluster

Increase the Number of Containers

During cluster usage, if more computational resources are needed, you can increase the number of containers at any time. Follow these steps:

1. Select the container cluster you want to scale out in the 【Elastic Training Cluster】 list
2. Click the [Scale Containers] button at the top

> **Notics**
>
> When scaling out, it is recommended to stop running scripts before performing the scaling configuration.

3. In the [Adjust cluster container count to] field, enter the new total number of containers.

> **Note**
>
> - This number indicates the total number and includes currently established worker containers and one launcher container.
> - The minimum number is 2
> - The system will automatically increase the number of worker containers and update the hostfile.

4. After confirming the number, click [Confirm].
5. Return to the [Containers] page to view the newly added containers.
6. When the container status updates to "Running," the cluster scaling and startup are complete.

Decrease the Number of Containers

When the training cluster no longer requires as many resources, you can remove some containers based on actual needs to release resources. Follow these steps:

1. Select the container cluster you want to scale down in the 【Elastic Training Cluster】 list.
2. Click the [Scale Containers] button at the top.

> **Notice**
>
> When scaling down, it is recommended to stop running scripts before performing the scaling configuration.

3. In the [Adjust cluster container count to] field, enter the new total number of containers.

> **Note**
>
> - This number indicates the total number and includes currently established worker containers and one launcher container.
> - The minimum number is 2.
> - The system will automatically decrease the number of worker containers and update the hostfile.

4. After confirming the number, click [Confirm].
5. Return to the [Containers] page and wait for the reduced containers to be removed from the list, indicating that cluster scaling is complete.

Delete Container Clusters

In the 【Elastic Training Cluster】 page, follow these steps to delete a container cluster:

1. Select the container cluster you want to delete (multi-selection supported).
2. Click the [Delete] button.
3. After the confirmation dialog appears, confirm that the selected containers are the ones you want to delete.
4. Click the [Delete] button again to complete the deletion.